

KITABU QUEST

S2

ICT



Cover scene

learners using computers safely with a mountain gorilla mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

2

Title Page

KITABU QUEST S2 ICT

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S2 learners study ICT one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Practise safe, useful digital skills and explain each step clearly.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Data protection
2. ICT in financial transactions
3. Objects in a document
4. Spreadsheet basics
5. Worksheet data presentation
6. Arc GIS
7. Complex formulae and functions
8. Network components and social media applications
9. Game programming

As you finish each topic, return to the success criteria and correct one answer before moving on.

Data protection: Lesson Opener

Learning focus: Data protection

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Data protection.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- use computers safely and securely to ensure that data is protected

Inquiry questions:

- How can data protection help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Data protection: Learn

Data protection is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Data protection
- Data
- protection
- idea
- skill
- example
- practice
- correction

Data protection: Worked Example

Worked example for Data protection: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called data-protection-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Data protection: Vocabulary And Study Skill

Use these words and study moves before you practise Data protection. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Data protection
- Data
- protection
- idea
- skill
- example
- practice
- correction

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Data protection without a clear example, method, or evidence.

Data protection: Guided Activity

Guided digital activity for Data protection:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Data protection: Practice And Correction

Digital practice for Data protection:

1. Complete a small digital-skills task for Data protection: use computers safely and securely to ensure that data is protected. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Data protection: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Data protection.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Data protection: Outcome Check

Outcome check for Data protection: Use computers safely and securely to ensure that data is protected.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

ICT in financial transactions: Lesson Opener

Learning focus: ICT in financial transactions

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for ICT in financial transactions.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- analyse and criticize the role and impact of computing tools in financial transactions. Use computing tools in financial transactions

Inquiry questions:

- How can ict in financial transactions help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

ICT in financial transactions: Learn

ICT in financial transactions is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- ICT in financial transactions
- financial
- transactions
- idea
- skill
- example
- practice
- correction

ICT in financial transactions: Worked Example

Worked example for ICT in financial transactions: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called ict-in-financial-transactions-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

ICT in financial transactions: Vocabulary And Study Skill

Use these words and study moves before you practise ICT in financial transactions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- ICT in financial transactions
- financial
- transactions
- idea
- skill
- example
- practice
- correction

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about ICT in financial transactions without a clear example, method, or evidence.

ICT in financial transactions: Guided Activity

Guided digital activity for ICT in financial transactions:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

ICT in financial transactions: Practice And Correction

Digital practice for ICT in financial transactions:

1. Complete a small digital-skills task for ICT in financial transactions: analyse and criticize the role and impact of computing tools in financial transactions. Use computing tools in financial transactions. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

ICT in financial transactions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from ICT in financial transactions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

ICT in financial transactions: Outcome Check

Outcome check for ICT in financial transactions: Analyse and criticize the role and impact of computing tools in financial transactions. Use computing tools in financial transactions.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Objects in a document: Lesson Opener

Learning focus: Objects in a document

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Objects in a document.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- insert symbols, pictures, tables and objects into a document

Inquiry questions:

- How can objects in a document help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Objects in a document: Learn

Objects in a document is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Objects in a document
- Objects
- document
- idea
- skill
- example
- practice
- correction

Objects in a document: Worked Example

Worked example for Objects in a document: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called objects-in-a-document-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Objects in a document: Vocabulary And Study Skill

Use these words and study moves before you practise Objects in a document. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Objects in a document
- Objects
- document
- idea
- skill
- example
- practice
- correction

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Objects in a document without a clear example, method, or evidence.

Objects in a document: Guided Activity

Guided digital activity for Objects in a document:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Objects in a document: Practice And Correction

Digital practice for Objects in a document:

1. Complete a small digital-skills task for Objects in a document: insert symbols, pictures, tables and objects into a document. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Objects in a document: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Objects in a document.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Objects in a document: Outcome Check

Outcome check for Objects in a document: Insert symbols, pictures, tables and objects into a document.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Spreadsheet basics: Lesson Opener

Learning focus: Spreadsheet basics

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Spreadsheet basics.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- work with a spreadsheet and apply basic manipulation of cell content using arithmetical operations

Inquiry questions:

- How can spreadsheet basics help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Spreadsheet basics: Learn

Spreadsheet basics is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Spreadsheet basics
- Spreadsheet
- basics
- idea
- skill
- example
- practice
- correction

Spreadsheet basics: Worked Example

Worked example for Spreadsheet basics: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called spreadsheet-basics-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Spreadsheet basics: Vocabulary And Study Skill

Use these words and study moves before you practise Spreadsheet basics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Spreadsheet basics
- Spreadsheet
- basics
- idea
- skill
- example
- practice
- correction

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Spreadsheet basics without a clear example, method, or evidence.

Spreadsheet basics: Guided Activity

Guided digital activity for Spreadsheet basics:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Spreadsheet basics: Practice And Correction

Digital practice for Spreadsheet basics:

1. Complete a small digital-skills task for Spreadsheet basics: work with a spreadsheet and apply basic manipulation of cell content using arithmetical operations. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Spreadsheet basics: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Spreadsheet basics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Spreadsheet basics: Outcome Check

Outcome check for Spreadsheet basics: Work with a spreadsheet and apply basic manipulation of cell content using arithmetical operations.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Worksheet data presentation: Lesson Opener

Learning focus: Worksheet data presentation

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Worksheet data presentation.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- manage a window, sorting and filtering data in a spreadsheet

Inquiry questions:

- How can worksheet data presentation help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Worksheet data presentation: Learn

Worksheet data presentation is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Worksheet data presentation
- Worksheet
- data
- presentation
- idea

Worksheet data presentation: Worked Example

Worked example for Worksheet data presentation: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called worksheet-data-presentation-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Worksheet data presentation: Vocabulary And Study Skill

Use these words and study moves before you practise Worksheet data presentation. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Worksheet data presentation
- Worksheet
- data
- presentation
- idea
- skill
- example
- practice

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Worksheet data presentation without a clear example, method, or evidence.

Worksheet data presentation: Guided Activity

Guided digital activity for Worksheet data presentation:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Worksheet data presentation: Practice And Correction

Digital practice for Worksheet data presentation:

1. Complete a small digital-skills task for Worksheet data presentation: manage a window, sorting and filtering data in a spreadsheet. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Worksheet data presentation: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Worksheet data presentation.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Worksheet data presentation: Outcome Check

Outcome check for Worksheet data presentation: Manage a window, sorting and filtering data in a spreadsheet.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Arc GIS: Lesson Opener

Learning focus: Arc GIS

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Arc GIS.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- fill a new empty map with data, use simple symbols, label features and attributes table, and navigate a map

Inquiry questions:

- How can arcgis help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Arc GIS: Learn

Arc GIS is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Arc GIS
- idea
- skill
- example
- practice
- correction

Arc GIS: Worked Example

Worked example for Arc GIS: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called arc-gis-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Arc GIS: Vocabulary And Study Skill

Use these words and study moves before you practise Arc GIS. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Arc GIS
- idea
- skill
- example
- practice
- correction

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Arc GIS without a clear example, method, or evidence.

Arc GIS: Guided Activity

Guided digital activity for Arc GIS:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Arc GIS: Practice And Correction

Digital practice for Arc GIS:

1. Complete a small digital-skills task for Arc GIS: fill a new empty map with data, use simple symbols, label features and attributes table, and navigate a map. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Arc GIS: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Arc GIS.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Arc GIS: Outcome Check

Outcome check for Arc GIS: Fill a new empty map with data, use simple symbols, label features and attributes table, and navigate a map.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Complex formulae and functions: Lesson Opener

Learning focus: Complex formulae and functions

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Complex formulae and functions.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- work with spreadsheets to apply complex formula and functions recognising the order of operations.
- Apply conditional formatting to the content of a worksheet. Use absolute and relative referencing

Inquiry questions:

- How can complex formulae and functions help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Complex formulae and functions: Learn

Complex formulae and functions is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Complex formulae and functions
- Complex
- formulae
- functions
- idea
- skill
- example
- practice

Complex formulae and functions: Worked Example

Worked example for Complex formulae and functions: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called complex-formulae-and-functions-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Complex formulae and functions: Vocabulary And Study Skill

Use these words and study moves before you practise Complex formulae and functions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Complex formulae and functions
- Complex
- formulae
- functions
- idea
- skill
- example
- practice

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Complex formulae and functions without a clear example, method, or evidence.

Complex formulae and functions: Guided Activity

Guided digital activity for Complex formulae and functions:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Complex formulae and functions: Practice And Correction

Digital practice for Complex formulae and functions:

1. Complete a small digital-skills task for Complex formulae and functions: work with spreadsheets to apply complex formula and functions recognising the order of operations. Apply conditional formatting to the content of a worksheet. Use absolute and relative referencing. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Complex formulae and functions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Complex formulae and functions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Complex formulae and functions: Outcome Check

Outcome check for Complex formulae and functions: Work with spreadsheets to apply complex formula and functions recognising the order of operations. Apply conditional formatting to the content of a worksheet. Use absolute and relative referencing.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Network components and social media applications: Lesson Opener

Learning focus: Network components and social media applications

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Network components and social media applications.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- identify computer network devices, medium and peripherals. Connect different computer devices to the network. Use social media to exchange information on social life

Inquiry questions:

- How can network components and social media applications help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Network components and social media applications: Learn

Network components and social media applications is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Network components and social media applications
- Network
- components
- social
- media
- applications
- idea
- skill

Network components and social media applications: Worked Example

Worked example for Network components and social media applications: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called network-components-and-social-media-applications-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Network components and social media applications: Vocabulary And Study Skill

Use these words and study moves before you practise Network components and social media applications. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Network components and social media applications
- Network
- components
- social
- media
- applications
- idea
- skill

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Network components and social media applications without a clear example, method, or evidence.

Network components and social media applications: Guided Activity

Guided digital activity for Network components and social media applications:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Network components and social media applications: Practice And Correction

Digital practice for Network components and social media applications:

1. Complete a small digital-skills task for Network components and social media applications: identify computer network devices, medium and peripherals. Connect different computer devices to the network. Use social media to exchange information on social life. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Network components and social media applications: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Network components and social media applications.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Network components and social media applications: Outcome Check

Outcome check for Network components and social media applications: Identify computer network devices, medium and peripherals. Connect different computer devices to the network. Use social media to exchange information on social life.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Game programming: Lesson Opener

Learning focus: Game programming

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Game programming.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- design a game, collect data in a table and add game rules

Inquiry questions:

- How can game programming help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Game programming: Learn

Game programming is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Game programming
- Game
- programming
- idea
- skill
- example
- practice
- correction

Game programming: Worked Example

Worked example for Game programming: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called game-programming-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Game programming: Vocabulary And Study Skill

Use these words and study moves before you practise Game programming. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Game programming
- Game
- programming
- idea
- skill
- example
- practice
- correction

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Game programming without a clear example, method, or evidence.

Game programming: Guided Activity

Guided digital activity for Game programming:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Game programming: Practice And Correction

Digital practice for Game programming:

1. Complete a small digital-skills task for Game programming: design a game, collect data in a table and add game rules. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Game programming: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Game programming.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Game programming: Outcome Check

Outcome check for Game programming: Design a game, collect data in a table and add game rules.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Data protection: Review Clinic 1

Use this review clinic to strengthen ICT without copying earlier answers.

1. Choose one idea from Data protection that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

ICT in financial transactions: Review Clinic 2

Use this review clinic to strengthen ICT without copying earlier answers.

1. Choose one idea from ICT in financial transactions that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Objects in a document: Review Clinic 3

Use this review clinic to strengthen ICT without copying earlier answers.

1. Choose one idea from Objects in a document that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Spreadsheet basics: Review Clinic 4

Use this review clinic to strengthen ICT without copying earlier answers.

1. Choose one idea from Spreadsheet basics that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Data protection: Application Workshop 1

Digital application workshop 1 for Data protection.

Grade move: connect ideas across topics and explain causes, methods, or consequences; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

ICT in financial transactions: Application Workshop 2

Digital application workshop 2 for ICT in financial transactions.

Grade move: connect ideas across topics and explain causes, methods, or consequences; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Glossary

Use this glossary to revise important words in ICT.

- Data protection: Explain this term using an example from the book, your school, or your community.
- ICT in financial transactions: Explain this term using an example from the book, your school, or your community.
- Objects in a document: Explain this term using an example from the book, your school, or your community.
- Spreadsheet basics: Explain this term using an example from the book, your school, or your community.
- Worksheet data presentation: Explain this term using an example from the book, your school, or your community.
- Arc GIS: Explain this term using an example from the book, your school, or your community.
- Complex formulae and functions: Explain this term using an example from the book, your school, or your community.
- Network components and social media applications: Explain this term using an example from the book, your school, or your community.
- Game programming: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong ICT answer responds to the exact task, uses correct vocabulary, gives a clear example or evidence, and shows correction after feedback. For practical work, name materials, safety, process, quality checks, peer feedback, and one improvement made after review.

Final Project

Create a final project for ICT that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.